

Oestrogen levels in serum and urine of premenopausal women eating low and high amounts of meat.

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OBJECTIVE: Based on the hypothesis that high-meat diets may increase breast cancer risk through hormonal pathways, the present analysis compared oestrogens in serum and urine by meat-eating status. **DESIGN:** Intervention with repeated measures. **SETTING:** Two randomized soya trials (BEAN1 and BEAN2) among premenopausal healthy women. **SUBJECTS:** BEAN1 participants completed seven unannounced 24 h dietary recalls and donated five blood and urine samples over 2 years. BEAN2 women provided seven recalls and three samples over 13 months. Serum samples were analysed for oestrone (E α) and oestradiol (E β) using RIA. Nine oestrogen metabolites were measured in urine by LC-MS. Semi-vegetarians included women who reported consuming <30 g of red meat, poultry and fish daily, and pescatarians those who reported consuming <20 g of meat/poultry but >10 g of fish daily. All other women were classified as non-vegetarians. We applied mixed models to compute least-square means by vegetarian status adjusted for potential confounders. **RESULTS:** The mean age of the 272 participants was 41.9 (SD 4.5) years. Serum E α (85 v. 100 pg/ml, $P = 0.04$) and E β (140 v. 154 pg/ml, $P = 0.04$) levels were lower in the thirty-seven semi-vegetarians than in the 235 non-vegetarians. The sum of the nine urinary oestrogen metabolites (183 v. 200 pmol/mg creatinine, $P = 0.27$) and the proportions of individual oestrogens and pathways did not differ by meat-eating status. Restricting the models to the samples collected during the luteal phase strengthened the associations. **CONCLUSIONS:** Given the limitations of the study, the lower levels of serum oestrogens in semi-vegetarians than non-vegetarians need confirmation in larger populations.